Pranav Sankar

Machine Learning & AI Engineer | Forecasting, NLP, Deep Learning | Agentic AI & MLOps

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PROFESSIONAL SUMMARY

Machine Learning and AI Engineer with 4 years of experience building production systems spanning time-series forecasting, statistical process monitoring, NLP, deep learning, model evaluation, generative AI, and agentic workflows. Experienced in taking solutions from data preparation and experimentation through Python APIs, asynchronous model execution, AWS deployment, evaluation, monitoring, and autoscaling. Combines applied ML depth with production software engineering and direct delivery across 20+ client engagements.

TECHNICAL SKILLS

Machine Learning: Time-series forecasting, SARIMA/SARIMAX, ARIMA, Monte Carlo simulation, PERT, statistical process control, classification, sentiment analysis, feature and data processing

Deep Learning & NLP: TensorFlow, PyTorch, CNN, LSTM, Swin-UNet, BERT, TextBlob, Qwen TTS, model fine-tuning

GenAI & Agentic AI: LangGraph, AutoGen, LangChain, RAG, vector databases, tool calling, human-in-the-loop, state persistence, LLM evaluation

MLOps & Cloud: Weights & Biases, AWS CloudWatch, ECS, Fargate, EC2, S3, CloudFront, Docker, RunPod, GitHub Actions CI/CD, model and artifact versioning

Engineering: Python, Django, Django REST Framework, FastAPI, Flask, pandas, NumPy, statsmodels, PostgreSQL, REST APIs, Django Q2

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Codersarts
Apr 2023 - Present

Production ML, AI product engineering, MLOps, and client delivery

- Built and deployed production ML and AI capabilities using Python, Django/DRF, FastAPI, pandas, NumPy, statsmodels, TensorFlow, PyTorch, LangGraph, and AWS.
- Implemented Compelion's project-level benefit forecasting with SARIMA/SARIMAX, ARIMA-based metric forecasting, PERT and 1,000-iteration Monte Carlo time/cost estimation, and statistical process control for outlier, shift, trend, and oscillation detection.
- Deployed Compelion ML workloads as Django Q2 background tasks on AWS; used CloudWatch monitoring to tune EC2 autoscaling for the full backend under client cost constraints.
- Developed enterprise NLP analytics over project comments and notes, including sentiment scoring, monthly and project-type aggregation, role-level breakdowns, and SPC-based sentiment trend monitoring exposed through Django REST APIs.
- Built a 60-email evaluation suite for Nova covering relevance gating, classification, sentiment, multilingual behavior, prompt injection, RAG grounding, deterministic reply checks, LLM judging, latency, and cost; retained GPT-4o-mini after candidate models showed material quality regressions.
- Built production LangGraph workflows for an NDA-covered analytics copilot and the Nova personal assistant using human-in-the-loop review, persisted state, tool calling, email and calendar actions, Vonage phone provisioning, and Vapi agent integrations.
- Tracked fine-tuning experiments, model artifacts, and versions with Weights & Biases; monitored AWS applications with CloudWatch and delivered containerized CI/CD across EC2, ECS/Fargate, and RunPod.
- Fine-tuned Qwen 3.5 TTS on the CEO's voice and deployed a serverless FastAPI endpoint on RunPod, reducing 12-minute audio generation to under 3 minutes.
- Implemented ambiguity checks, document-ingestion pipelines, prompt-injection defenses, and role-based access control for the Agents platform.
- Served as the primary technical contact for 20+ clients across discovery, model and architecture decisions, demonstrations, delivery, and production rollout.

Machine Learning Engineer Intern | Codersarts
Aug 2022 - Mar 2023

Applied ML, deep-learning research, and NLP foundations

- Developed and evaluated machine learning, computer vision, and NLP prototypes using TensorFlow and PyTorch, including classification and sentiment-analysis pipelines.
- Supported client research by delivering a trained CNN-LSTM model for ECG disease classification and fine-tuning a Swin-UNet model for X-ray image classification on client datasets.

SELECTED ML & AI ENGINEERING WORK

Compelion | Forecasting, Risk Estimation & Statistical Intelligence

Engineered production forecasting and decision-support modules for an enterprise innovation platform: SARIMA/SARIMAX benefit forecasts, auto-ARIMA metric forecasts, PERT-based project estimates, 1,000-run Monte Carlo time/cost simulations, and six-rule SPC monitoring. Integrated results with Django models through transactional bulk create/update operations and asynchronous Django Q2 jobs.

Nova | Model Evaluation, NLP & Production AI Assistant

Created an evaluation pipeline that measures gating, classification, sentiment, language, RAG retrieval, grounded response quality, prompt-injection behavior, completion, latency, and cost. Benchmarked GPT-5-nano and GPT-5.4-nano against the GPT-4o-mini baseline and used regression evidence to retain the more reliable production model. Also prototyped BERT-based email classification and sentiment analysis before rejecting production adoption on cost grounds.

NDA-Covered Analytics Copilot | Production LangGraph Assistant

Developed a data-backed chatbot embedded in a client analytics product. Built LangGraph orchestration with persisted state and human-in-the-loop controls to answer natural-language analytics questions and generate chart-driven insights; deployed the workflow to production.

Applied Deep Learning | Healthcare Research Assistance

Delivered a CNN-LSTM disease-classification model for ECG sequences and fine-tuned Swin-UNet for X-ray image classification using client-provided datasets. Applied data preparation, training, evaluation, and model-delivery workflows with TensorFlow and PyTorch.

Agents Platform | Responsible and Secure AI Controls

Built reusable agentic workflows with document ingestion, ambiguity detection, prompt-injection checks, and role-based access control. Combined LangGraph, AutoGen, RAG, vector retrieval, and human oversight patterns for production-oriented business automation.

CloudNotes & Qwen TTS | ML Deployment & Scaling

Owned production deployment for an audio-to-structured-report platform using ECS/Fargate autoscaling, S3, CloudFront, and GitHub Actions. Separately fine-tuned Qwen 3.5 TTS for internal voice automation and served the model through a modular FastAPI endpoint on RunPod.

MLOPS & MODEL GOVERNANCE

- Experiment tracking, artifact management, and versioning with Weights & Biases
- CloudWatch logs, metrics, operational monitoring, and autoscaling decisions
- Task-level and pipeline-level evaluation for ML, RAG, LLM, and agent workflows
- Quality, latency, and cost benchmarking before production model changes
- Prompt-injection checks, RBAC, ambiguity handling, and human review

ADDITIONAL PRODUCTION WORK

- **Candidate Hub:** Semantic resume-JD matching with embeddings, contextual scoring, and skill-gap reasoning
- **DocProcessing360:** Vision-language document extraction and production rollout support
- **Agents (CaaS):** Reusable chatbot and agent automation platform
- **Quizentia:** AI assessment generation; product and deployment ownership

EDUCATION

M.Tech, Digital Manufacturing

SASTRA University | 2020 - 2022 | CGPA: 8.76

B.Tech, Mechanical Engineering

SRM Institute of Science and Technology | 2015 - 2019 | CGPA: 6.96

SELECTED CERTIFICATIONS

- MLOps Specialization (GCP), DeepLearning.AI
- IBM Full Stack Software Developer
- Docker, Kubernetes and OpenShift
- Microservices and Serverless Development